Questions by topic: Ionising radiation - Set 1 (60 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows some radioactive isotopes, their half-lives and the type of radiation they emit.

Table 1

Isotope	Half-life	Main radiation	Source
Iodine-131	8 days	Beta / gamma	Fission in reactors
Strontium-90	29 years	Beta	Fission in reactors
Caesium-137	30 years	Beta / gamma	Fission in reactors
Plutonium-239	24 400 years	Alpha	Neutron bombardment of U-238
Uranium-238	4.5×10^9 years	Alpha	Occurs naturally in rocks

0 1 . 1 Define the term **half-life**.

[1 mark]

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0 1 . 2 A sample of iodine-131 has an activity of 6400 Bq. Calculate its activity after 32 days.

[2 marks]

Answer _____ Bq

0 1 . 3 Use Table 1 to explain why isotopes with half-lives of several decades, such as caesium-137, cause the greatest concern after a nuclear accident.

[3 marks]

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0 1 . 4 Explain why plutonium-239 is relatively safe outside the body but very dangerous if inhaled or ingested.

[3 marks]

0 1 . 5 Explain what is meant by an **activation product**.

[1 mark]

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0 2 . 1 Explain how ionising radiation damages living cells.

[3 marks]

0 2 . 2 Distinguish between acute and chronic effects of radiation exposure.

[2 marks]

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0 2 . 3 Distinguish between **exposure** and **contamination**.

[2 marks]

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0 2 . 4 A worker receives an absorbed dose of 0.002 Gy from alpha radiation.
Calculate the effective dose in sieverts.

[2 marks]

Answer ..... Sv

0 2 . 5 State what is measured by the becquerel.

[1 mark]

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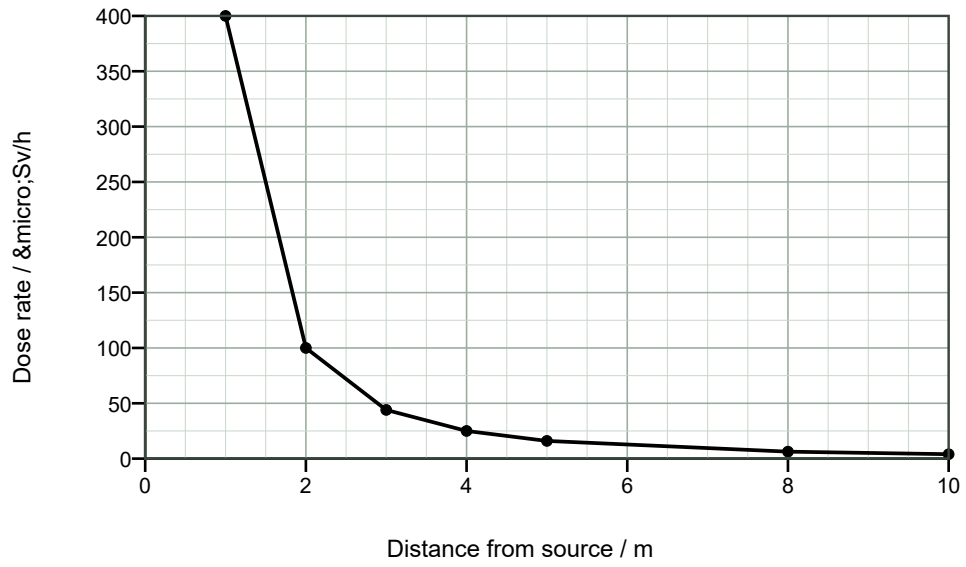
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0 3 Figure 1 shows the radiation dose rate measured at different distances from a small gamma source.

Figure 1



0 3 . 1 Use Figure 1 to show that the dose rate follows the inverse square law.

[2 marks]

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0 3 . 2 Calculate the dose rate at 20 m from the source.

[1 mark]

Answer µSv/h

0 3 . 3 Explain how each of the following reduces the exposure of workers handling radioactive sources.

[4 marks]

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Closed (sealed) sources .....

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Absorbing materials .....

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Reduced exposure time .....

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Worker monitoring .....

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0 3 . 4 Explain why potassium iodide tablets are issued to people near a nuclear accident.

[2 marks]

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0 3 . 5 State the meaning of ALARA.

[1 mark]

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0 4 Table 1 shows the average annual radiation dose received by a person in the UK from different sources.

Table 1

Source	Percentage of average dose / %
Radon gas from the ground	50
Medical (X-rays, CT scans)	16
Gamma rays from rocks / buildings	13
Internal (food and drink)	9.5
Cosmic radiation	10
Occupational, fallout, nuclear discharges, consumer products	1.5

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0 4 . 1 Explain why radon concentrations are high inside some houses in Devon and Cornwall.

[3 marks]

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0 4 . 2 Use Table 1 to calculate the percentage of the average dose that comes from artificial (human-made) sources.

[1 mark]

Answer ..... %

0 4 . 3 Suggest **two** reasons why the dose received by an individual may differ greatly from the average.

[2 marks]

1 .....

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2 .....

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0 4 . 4 Explain why a risk:benefit analysis is needed for the use of X-rays in medicine, and give **two** reasons why the risks of ionising radiation are difficult to assess.

[4 marks]

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0 5 . 1 Complete Table 1 to show the source and disposal method for each category of radioactive waste from nuclear electricity generation.

[6 marks]

Table 1

Waste category	Example / source	Storage or disposal method	
High-level waste			
Intermediate-level waste			
Low-level solid waste			

0 5 . 2 Explain why high-level waste must be cooled and why its storage must be monitored for very long periods.

[2 marks]

0 5 . 3 Describe how low-level liquid and gaseous wastes are treated before discharge.

[2 marks]

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0 6 . 1 Evaluate the risks and benefits of the uses of ionising radiation, and explain the principles used to control exposure of workers and the public.

[9 marks]

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0 6 . 2 Give **one** reason why the risks and benefits of nuclear electricity may not fall on the same people.

[1 mark]

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END OF QUESTIONS

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